

Smart Home Energy Dashboard

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Course: ES115 – Engineering Computer Applications

Final Project Bank, Question 1

Date: May 6, 2026

Problem Overview / System Description

- What system are you modeling?

We are modeling a system that allows us to analyze home energy usage patterns and identify peak loads.

- Why is this system important in engineering?

It allows us to reduce cost by helping to reduce peak loads, which extends the life of electrical infrastructure.

- Real-world application (physics or chemistry context): A dashboard like this could be used in the real-world to monitor the heat flux of a home and inform the user on how to reduce their energy waste.

Engineering Objective

- What are you trying to analyze or predict?

We are trying to analyze what contributes most to the to a household's total energy expenditure, which allows the user to decrease their usage.

- What question does your model answer?

How much energy can we save by changing the consumers behavior?

Mathematical Model

- Define all variables and constants:
 - Main Table:
 - Days of observation
 - Power usage
 - MATLAB Code:
 - T ~ the array form of the table
 - K ~ a value used to denote the week of the study
 - N ~ a value used to denote the day of the study
 - X ~ a value used in creation of arrays for just weekends and weekdays

Algorithm / System Logic

- Each part of the question has a section of code to answer it.
- Part 1 – uploads data from an excel file
- Part 2 – determines averages and high values per week of power usage
- Part 3 – creates plots showing the data:
 - One time-series plot of energy usage
 - One histogram plot of energy usage
 - One histogram plot of weekend energy usage
 - One histogram plot of weekday energy usage
- Part 4 – discussed what changes can be made to decrease consumption

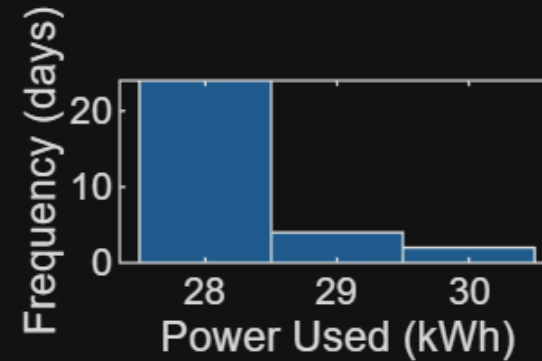
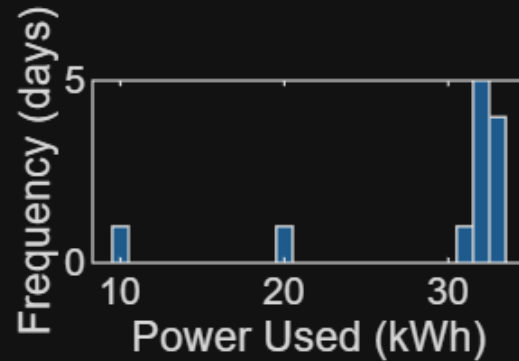
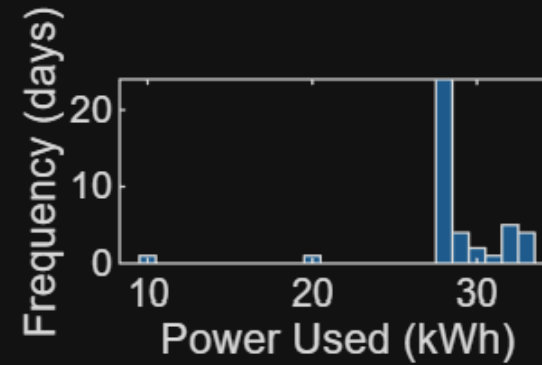
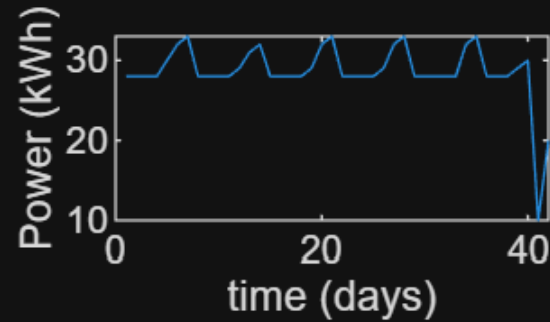
MATLAB Implementation

```
WeekEnd = [];% creating an array to separate out the weekend usage values from the table
x = 1; % placement variable for placing each value in the new array
while x <= 12 % while the array is smaller than its end size of 12
    k = 1;% value to denote which week data is being pulled from
    while k <= 6 % while the week value is less than or equal to six
        n = 1; % value to denote which day data is being pulled from
        while n >= 0 % while the day value is greater than zero, with a total of two values
            WeekEnd(x) = T(k*7-n,2); % pull data from the original array
            n = n - 1; % decrease day value by one
            x = x + 1; % increase placement values by one
        end % end day loop
        k = k + 1; % increase week value by one
    end % end week loop
end % end WeekDay loop

WeekDay = [];% creating an array to separate out the weekday usage values from the table
x = 1; % placement variable for placing each value in the new array
while x <= 12 % while the array is smaller than its end size of 30
    k = 1;% value to denote which week data is being pulled from
    while k <= 6 % while the week value is less than or equal to six
        n = 6; % value to denote which day data is being pulled from
        while n >= 2 % while the day value is greater than two, with a total of seven values
            WeekDay(x) = T(k*7-n,2); % pull data from the original array
            n = n - 1; % decrease day value by one
            x = x + 1; % increase placement values by one
        end % end day loop
        k = k + 1; % increase week value by one
    end % end week loop
end % end WeekDay loop
```

Simulation Results (Plots)

Power Usage Summary:



Model Limitations

- What assumptions did you make?

The assumptions made while creating the model, were the start day of the model; it was assumed that it started on a Monday. While this was accurate because the data was fabricated, when working with real measurements this assumption cannot be made.

- What does your model NOT capture?

The model only covers six weeks of energy consumption, therefore does not model changes in energy consumption due to weather changes. This could be rectified by taking a larger sample size that encompasses more than one season.

Engineering Relevance

- Why does this matter in real engineering?

It allows us to reduce cost by helping to reduce peak loads, which extends the life of electrical infrastructure.

- Where could this model be used?

It could be used in a home, school, library, residential building etc. It can be used anywhere that uses electricity.

Team Contributions

- What did each member do?
 - Annie: MATLAB coding, Analysis & visualization
 - Samuel: Model Development

Conclusion

- Summarize:
 - A mathematical model to analyze power consumption data from a house was created.
 - The end result showed weekend power consumption was greater than weekday power consumption
 - Key takeaways about engineering modeling:
 - There are many factors that can affect the outcome of a model
 - There are ways to present data that are more effective than others